

Trained



train



Large labelled training set

Environment-specific features

A

Standard



Cross-env.



Training-free, prior work



Frozen patch features

→



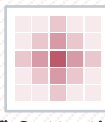
Codebook

or



Segmentation CLS attention

or



Prior-selected / aggregated features

B

Standard



Cross-env.

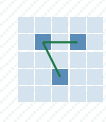


Ours

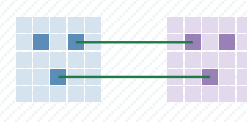


Frozen patch features

→



Within-image
Balanced, matchable patch evidence



Across-image
Balanced, matchable patch evidence

C

Standard

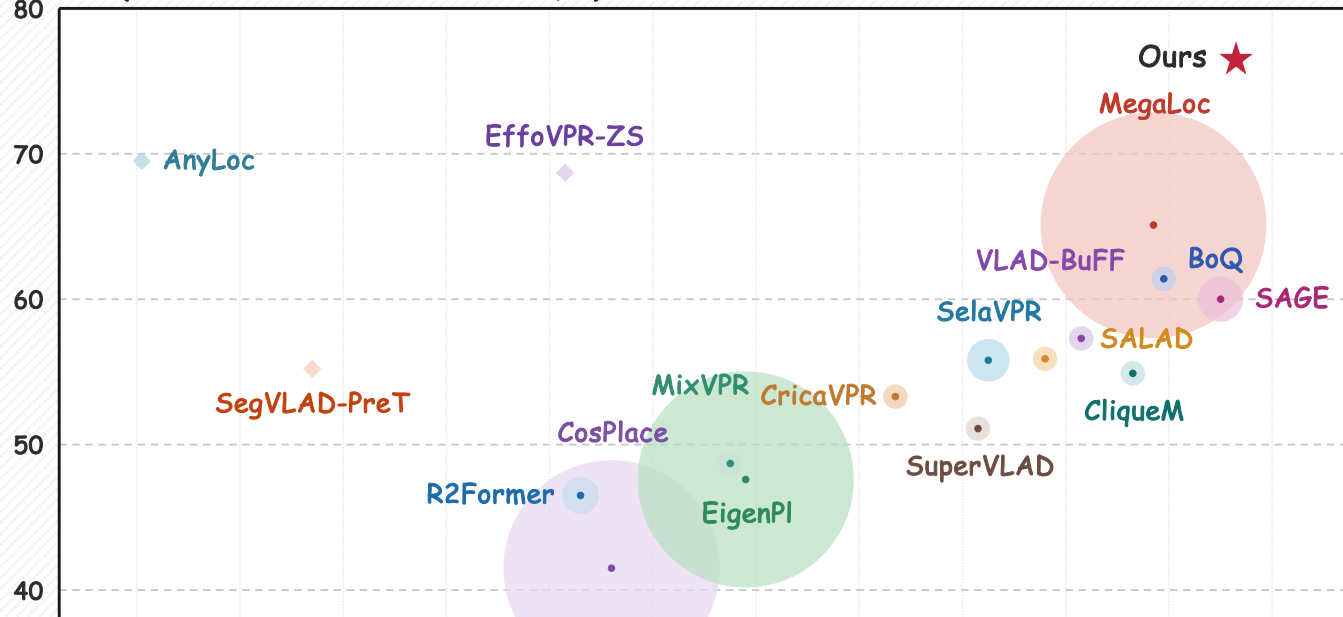


Cross-env.



Training cost vs. performance

mean R@1 (9 cross-environment benchmarks, %)



D

★ ours (training-free)

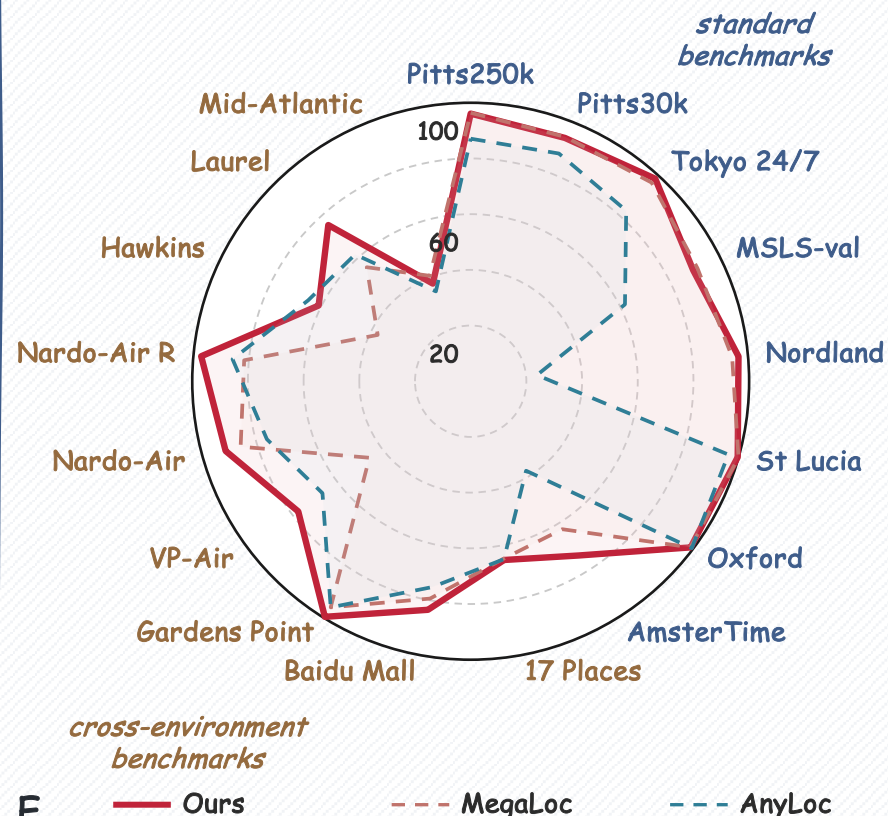
♦ training-free

● trained

● 1 million images

● 5 million images

Performance on 17 benchmarks



E

— Ours

- - - MegaLoc

- - - AnyLoc